

Back to Group Theory

Arham Lodha

November 04, 2023

Problem 1. We let finite G act on itself by conjugation. Consider an element g in G which IS NOT in the center. Write $Z(g)$ for its centralizer (the elements of G that commute with g , note they form a subgroup, $Z(g) = G^g$ for G acting on itself by conjugation). Explain why $|Z(G)|$ properly divides $|Z(g)|$ which properly divides $|G|$, where properly divides means it divides with quotient $\neq 1$ (here $Z(G)$ is the center of G). Use this to show that if $|G| = pq$ with p, q prime. Then either G is Abelian, or its center is trivial.

Solution. Suppose $g \notin Z(G)$. Both $Z(G)$ and $Z(g)$ are subgroups of G . $\forall h \in Z(G)$, $hgh^{-1} = g$ thus $h \in Z(g)$ since h commutes with all element of G by definition. $g^{-1}gg = g$ thus $g^{-1} \in Z(g)$ but since $g \notin Z(G) \implies g^{-1} \notin Z(G)$. Thus $Z(G) \subset Z(g)$. Thus $|Z(G)| \mid |Z(g)|$ where $|Z(g)| = k|Z(G)|$ where $k \in \mathbb{Z}^+$ and $k \neq 1$. Furthermore since $g \notin Z(G)$, then $Z(g) \subset G$. Thus $|Z(G)| \mid |G|$ where $|G| = m|Z(g)|$ where $m \in \mathbb{Z}^+$ and $m \neq 1$.

Suppose $|G| = pq$ where p, q are prime. Then the sizes of its subgroup is $1, p, q, pq$. There either $\exists g \notin Z(G)$ or $\forall g \in G, g \in Z(G)$. If it is the latter then G is Abelian. If it is 1st option. Then there exists $g \notin Z(G)$. Subgroup $Z(g)$ must have prime order. Moreover $Z(G) \subset Z(g)$, where $|Z(g)| = k|Z(G)|$ where $k \geq 1$. But since $|Z(g)|$ is prime its only factors are 1 and $|Z(g)|$ and since $k \neq 1$ then $k = |Z(g)|$ then $|Z(G)| = 1$ and the center is trivial. ■

Lemma . $\forall g \notin Z(G), |C_g| = \frac{|G|}{k|Z(G)|}$ for $k \in \mathbb{Z} \ni k > 1$

Proof. $\forall g \in G, |G| = |Z(g)||C_g| = k|Z(G)||C_g|$ by problem 1 for $k > 1$. Thus $|C_g| = \frac{|G|}{k|Z(G)|}$ for $k \in \mathbb{Z} \ni k > 1$ □

Problem 7.1.2. Let H be a subgroup of a group G . Describe the orbits for the operation of H on G by left multiplication.

Proof. Right cosets. $\forall g \in G, H \cdot g = \{hg : h \in H\} = Hg$ (all equalities are by definition). □

Problem 7.2.1a. Determine the centralizer and the order of the conjugacy class of $A = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix} \in GL_2(F_3)$

Solution. Suppose $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in Z(A)$. Then $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Thus $a = a + c$, $a + b = b + d$, $c + d = d$. Thus $c = 0$, $a = d$, and b is free.

Thus $Z(A) = \left\{ \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} : \forall a, b \in F_3 \right\}$. $|Z(A)| = 4$. Order of Conjugacy class is $|GL_2(F_3)| = 48/4 = 12$. ■

Problem 7.2.1b. Determine the centralizer and the order of the conjugacy class of $A = \begin{bmatrix} 1 & \\ & 2 \end{bmatrix} \in GL_2(F_5)$

Solution. Suppose $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in Z(A)$. Then $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & \\ & 2 \end{bmatrix} = \begin{bmatrix} a & 2b \\ c & 2d \end{bmatrix} = \begin{bmatrix} a & b \\ 2c & 2d \end{bmatrix} = \begin{bmatrix} 1 & \\ & 2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Thus $2b = b$, $2c = c$. Thus $b = c = 0$.

Thus $Z(A) = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} : \forall a, b \in F_5 \right\}$. $|Z(A)| = 16$. Order of Conjugacy class is $|GL_2(F_5)| = 480/16 = 30$. ■

Problem 7.2.6. Determine the conjugacy classes in the group M of isometries of the plane

Solution. Everything in M is either a T_v (translation), $\rho_{P,\theta}$ (rotation of θ about a point P), r_L reflection about a line, or $T_v \circ r_L$ (glide reflections). Let $D : M \rightarrow O(2)$ be the derivative homomorphism. $\forall \alpha \in M$, $\alpha T_v \alpha^{-1} = T_{D(\alpha)(v)}$. Since length has to be preserved all translations where by w where $|w| = |v|$. Circles on $\mathbb{R}^2$.

$\alpha \rho_{P,\theta} \alpha^{-1} = \alpha T_P \circ \rho_{0,\theta} \alpha^{-1} = T_{D(\alpha)(P)} \circ \rho_{0,\pm\theta} = \rho_{D(\alpha)(P) \pm \theta}$. So conjugacy class of $\rho_{P,\theta}$ are all rotations of $\pm\theta$.

$\alpha r_L \alpha^{-1} = r_{\alpha(L)}$. Thus all reflections are conjugate.

$\alpha T_v \circ r_L \alpha^{-1} = \alpha T_v \alpha^{-1} \circ \alpha r_L \alpha^{-1} = T_{D(\alpha)(v)} \circ r_{\alpha(L)}$. Thus the conjugacy class of $T_v \circ r_L$ is all $T_w \circ r_{L'}$ where $w \ni |w| = |v|$ and any line L' . ■

Problem 7.2.7a. Rule out as many as you can, as class equations for a group of order 10: $1 + 1 + 1 + 2 + 5$

Proof. Because there are 3 conjugacy classes with 1 element, $|Z(G)| = 3$. But $3 \nmid 10$. Thus $Z(G)$ is not a valid subgroup which is a contradiction and thus class equation is not valid either. □

Problem 7.2.7b. Rule out as many as you can, as class equations for a group of order 10: $1 + 2 + 2 + 5$

Proof. $D_5 = \langle x, y \mid x^5 = y^2 = e, xy = yx^{-1} \rangle$, where $x = \rho_{2\pi/5}$ and $y = r_{y=0}$. The conjugacy class of reflections is 5, because $x^a r_L x^{a-1} = r_{x(L)}$. x^1 conjugate to x^{-1} and x^2 is conjugate to x^{-2} because $r_L \rho_\theta r_L = \rho_\theta^{-1}$. The identity is conjugate to itself. Class equation for D_5 is $1 + 2 + 2 + 5$. Thus the class equation is valid. □

Problem 7.2.7c. Rule out as many as you can, as class equations for a group of order 10: $1 + 2 + 3 + 4$

Proof. Conjugacy class must divide the order of group and 3 and 4 don't divide 10. Thus class equation is invalid. □

Problem 7.2.7d. Rule out as many as you can, as class equations for a group of order 10: $1 + 1 + 2 + 2 + 2 + 2$

Proof. G has the class equation $1 + 1 + 2 + 2 + 2 + 2$.

$$Z(G) \cong \mu_2$$

By the Lemma, $10 = k(2)(2) = 4k$ for $k \in \mathbb{N}, k > 1$. This is a contradiction because $4 \nmid 10$. Thus the class equation is invalid. □

Problem 7.2.8a. Determine the possible class equations of nonabelian groups of order 8

Solution. The class equation can use 1, 2, or 4. $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$ describes an abelian group so not valid.

Class equation has 1 one. But then $2n + 4m = 7$ for $n, m \in \mathbb{N}$ which isn't possible.

Class equation has 2 ones. Then $1 + 1 + 2 + 4$ is not valid because $8 = k2(4) = 8k$ where $k > 1$ which is a contradiction. $1 + 1 + 2 + 2 + 2$ is the class equation for Q_8 and thus is valid.

Class equation can't have 3 ones because then $Z(G) \nmid G$.

Class equation has greater than 4 ones. This fails because of the lemma, as $8 = 4k|C_g|$ where $k > 1$.

Thus the only class equation for nonabelian groups of order 8 is $1 + 1 + 2 + 2 + 2$. ■

Problem 7.2.8b. Determine the possible class equations of nonabelian groups of order 21

Solution. Class equation is a sum of 1, 3, and 7.

Center has order 7. Then by lemma. $3 = k|C_g|$ where $k > 1$ and $|C_g| = 3, 7$. This is impossible.

Center has order 3. Then by the lemma, $7 = k|C_g|$ where $k > 1$ and $|C_g| = 3, 7$. Again this is impossible.

Center has order 1. $1 + 3 + 3 + 7 + 7$, is the only choice. ■

Problem 7.2.9a. Determine the class equation for Q_8

Proof. $Z(Q_8) = \{1, -1\}$. $Z(i) = \{\pm 1, \pm i\}$. $Z(j) = \{\pm 1, \pm j\}$. $Z(k) = \{\pm 1, \pm k\}$. Thus the class equation is $1 + 1 + 2 + 2 + 2 + 2$. □

Problem 7.2.9b. Determine the class equation for D_5

Proof. We have shown this in Problem 7.2.7b. Class equation is $1 + 2 + 2 + 5$ □

Problem 7.2.9c. Determine the class equation for Subgroup of $GL_2(F_3)$ of invertible upper triangular matrices.

Proof. A is this subgroup. $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \in A$, a has 2 options, b has 3 options, and c has 2 options. Thus $|A| = 12$. Possible Sizes of conjugacy classes 1, 2, 3, 4, 6.

$$Z(A) = \lambda I \text{ where } \lambda = 1, 2.$$

4 and 6 are not possible because of the lemma.

$Z(\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix})$ is all diagonal matrices and there are 4 of those.

$$\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \in Z\left(\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\right). \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} a & a+2b \\ 0 & 2c \end{bmatrix} = \begin{bmatrix} a & b+c \\ 0 & 2c \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}. \quad a+2b = b+c$$

thus $a+b=c$. $|Z(\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix})| = 4$

By using the fact that conjugacy class times centralizer is the whole group we see our partial class equation is $1 + 1 + 3 + 3$ and we have 4 more elements. However since the conjugacy classes cannot have 4 elements and the centralizer is only 2 elements the only way to get the four elements is to have 2 more conjugacy classes of size 2.

Thus the class equation is $1 + 1 + 2 + 2 + 3 + 3$. □